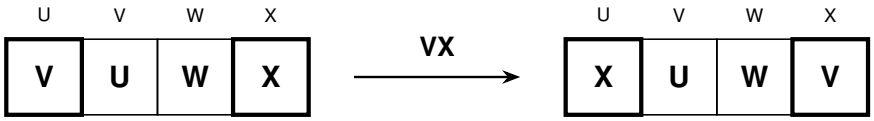
The homes stay in place. A card is at home when it sits in the home with its own letter. A swap trades the places of any two cards, near or far. Write each swap as the letters of the two cards it trades.

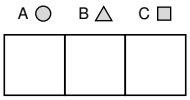


Problem 1: Start with every card at home. Reach a row with no card at home in as few swaps as you can, with three cards, with four and with all five. For five cards, explain why no fewer swaps will do.

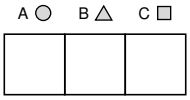
fewest swaps

three

swaps _____

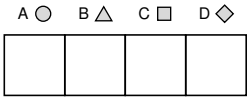


swaps _____

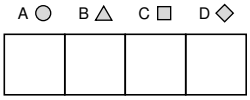


four

swaps _____

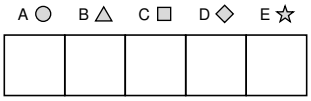


swaps _____

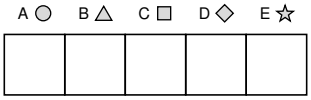


five

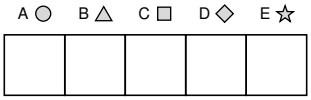
swaps _____



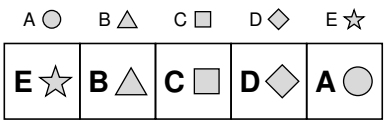
swaps _____



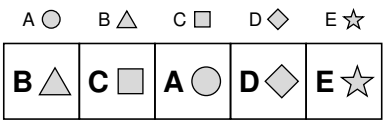
swaps _____



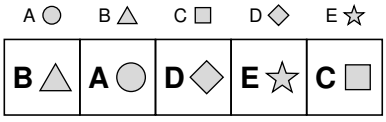
Problem 2: Set out each row below. Bring every card home using as few swaps as you can.



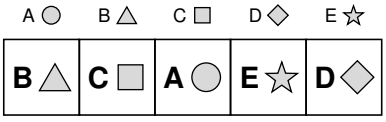
swaps _____ fewest ☐



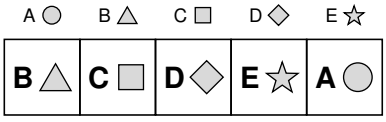
swaps _____ fewest ☐



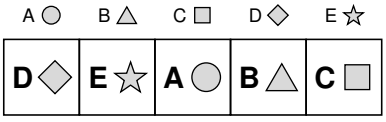
swaps _____ fewest ☐



swaps _____ fewest ☐



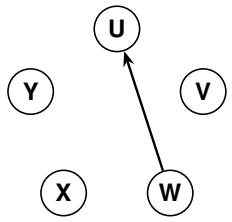
swaps _____ fewest ☐



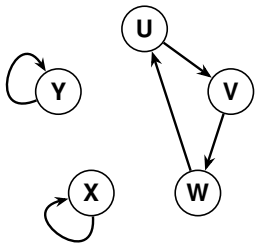
swaps _____ fewest ☐

Problem 3: An arrow goes from each card's letter to the letter of the home where the card sits.

U	V	W	X	Y
W	U	V	X	Y



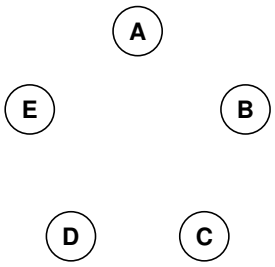
W sits in home U



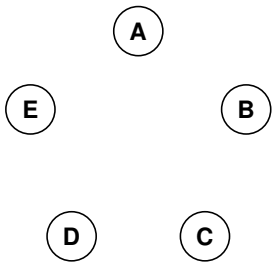
X and Y are at home

Draw the arrows for each row of Problem 2. How can you tell the fewest swaps from the arrows? Explain why no row can come home in fewer swaps than that.

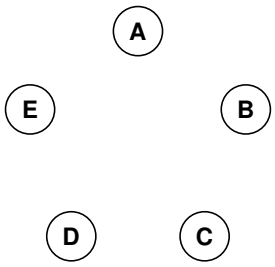
EBCDA



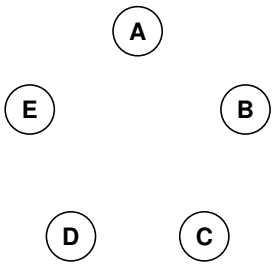
BCADE



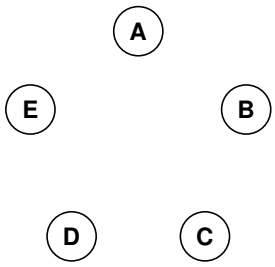
BADEC



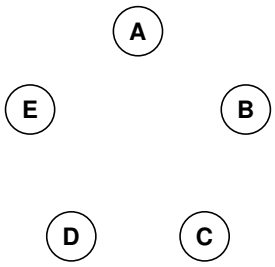
BCAED



BCDEA

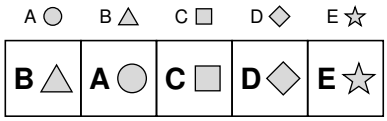


DEABC



Problem 4: Which rows of A, B, C, D and E need the most swaps to come home? How many such rows are there? How many of the 44 rows with no card at home need fewer swaps than that?

Problem 5: Can BACDE be brought home in exactly 2 swaps? In exactly 3, 4 or 5? Try the same numbers with BCADE. Explain which numbers of swaps can bring a row home.

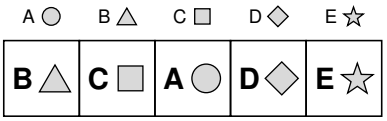


swaps _____

swaps _____

swaps _____

swaps _____



swaps _____

swaps _____

swaps _____

swaps _____

Problem 6: On the uncut sheet of 24 A–D outcome cards, circle every letter that sits in its own home. How many circles are there in all, and how many for each letter? Without listing, find how many circles the 120 rows of A, B, C, D and E would get, and explain.

A–D circles ☐ **A** ○ ☐ **B** △ ☐ **C** □ ☐ **D** ◇ ☐ A–E ☐

Problem 7: Any two circles on the same outcome card make a pair. How many pairs are on the 24 A–D outcome cards? Without listing, how many pairs would the 120 rows of A–E have?

A–D pairs ☐ A–E pairs ☐

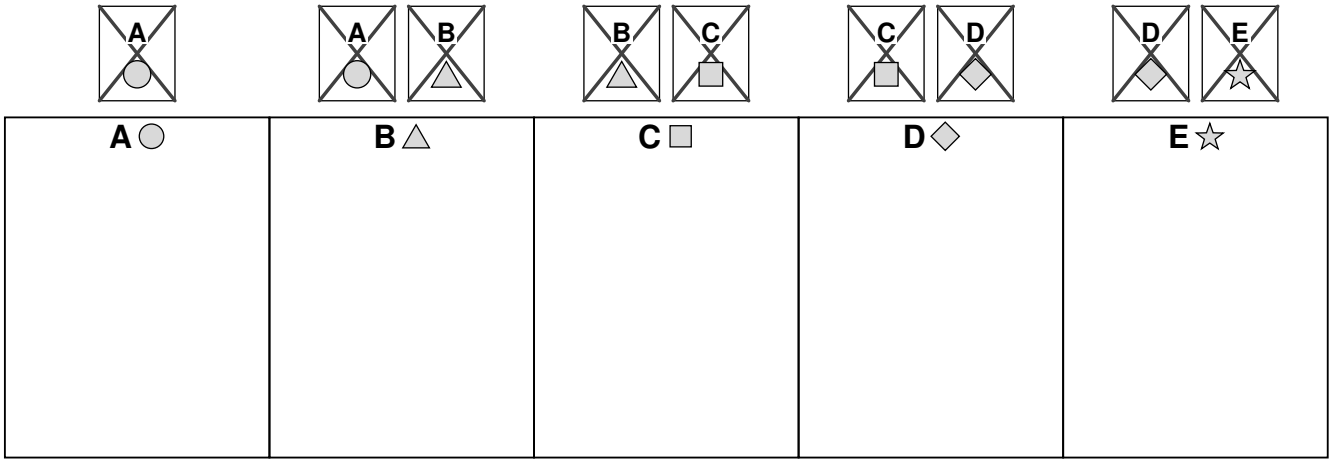
Problem 8: Sort the 24 cut A–D outcome cards into piles by how many letters are at home. Without listing, predict the piles for the 120 rows of A–E, and explain.

at home	**0**	**1**	**2**	**3**	**4**
A–D					

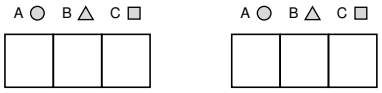
at home	**0**	**1**	**2**	**3**	**4**	**5**
A–E						

Problem 9: For three cards, the pile with no card at home has 2 rows and the pile with exactly one card at home has 3. Compare those two piles for four cards and for five cards. What happens, and why?

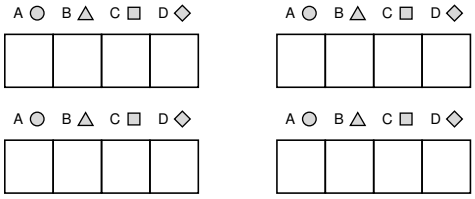
Problem 10: A card may not sit in a home that shows its picture crossed out. Find every row that obeys this board using homes A to C, using homes A to D, and using all five homes.



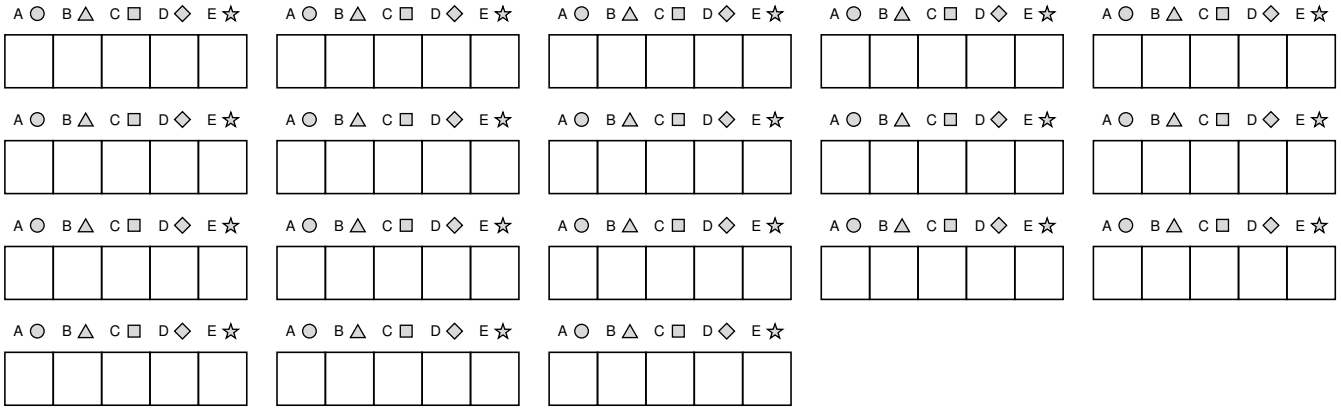
A to C



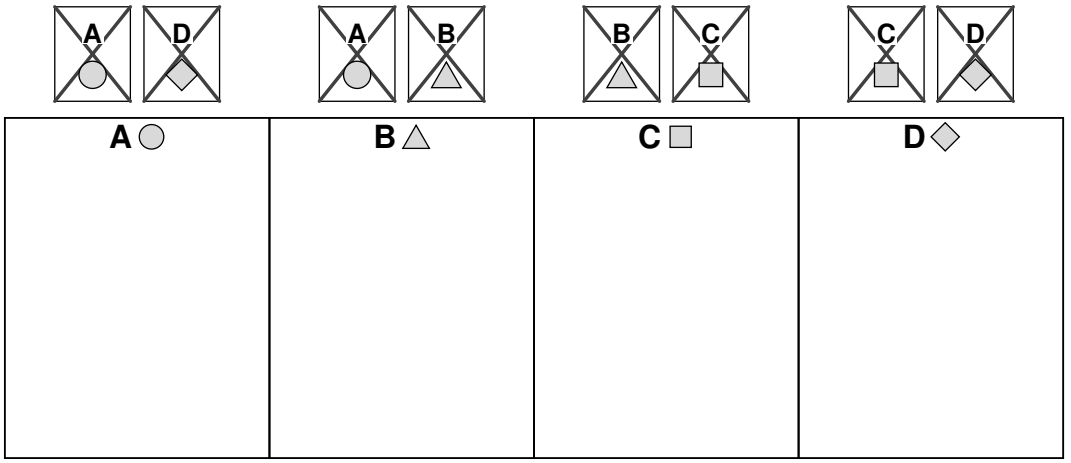
A to D

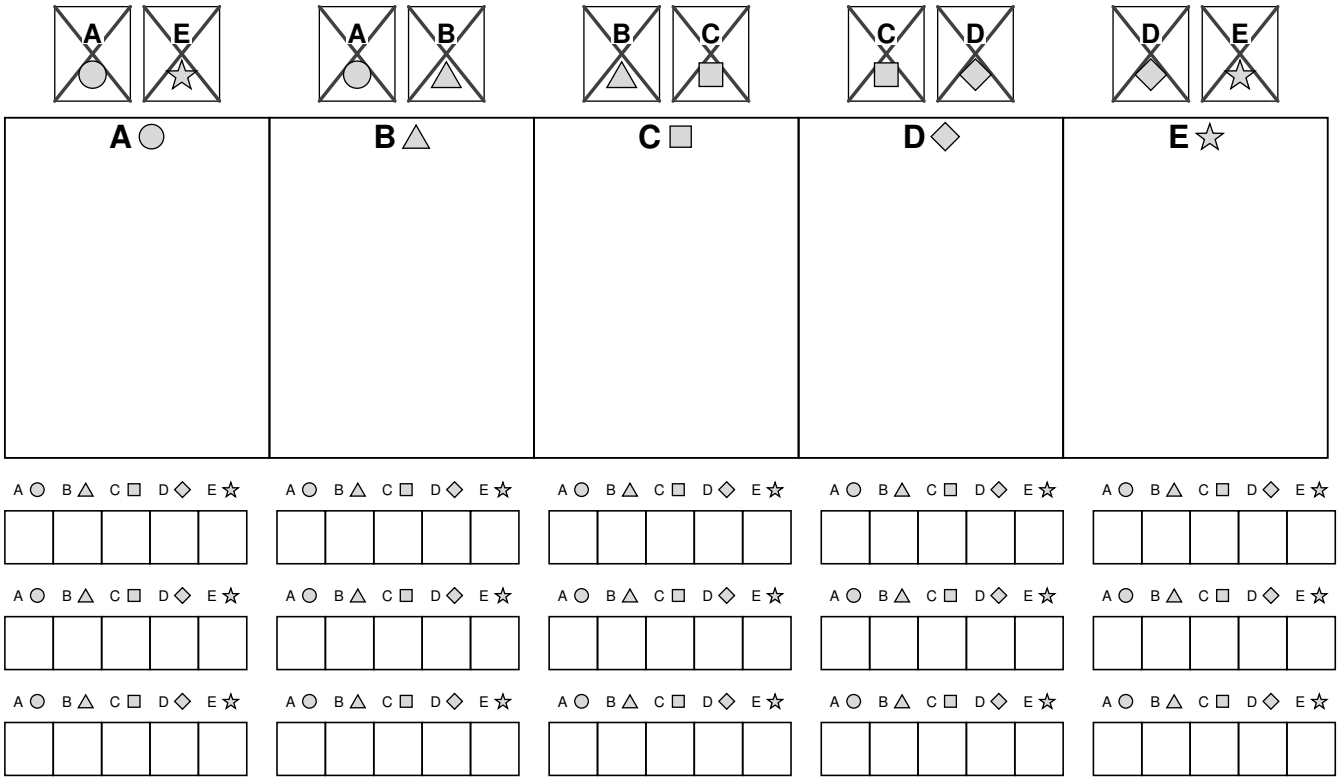


A to E

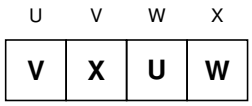


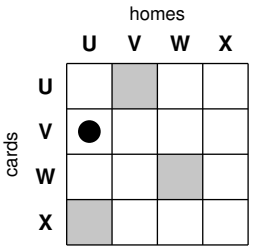
Problem 11: Find every row that obeys each of these boards.



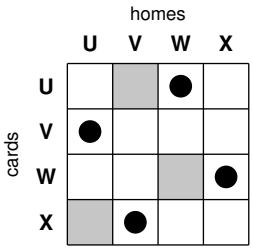


Problem 12: In a grid, a shaded square means that card may not sit in that home. A row puts one dot in every line across and every column.

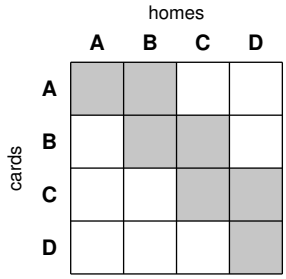




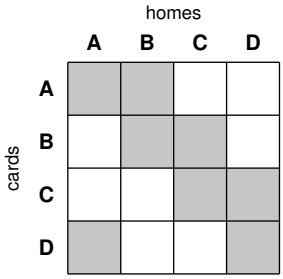
V sits in home U



Count the rows that obey each of these four-card grids without listing them.

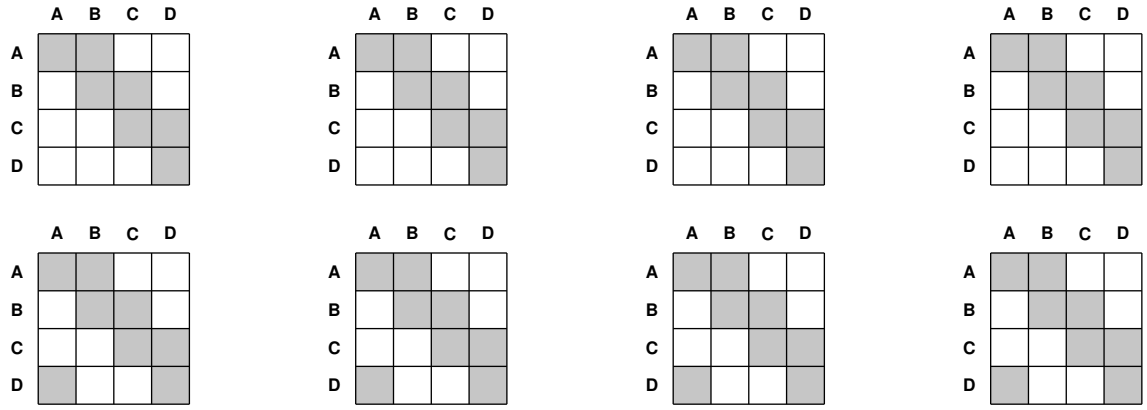


Problem 10, four cards

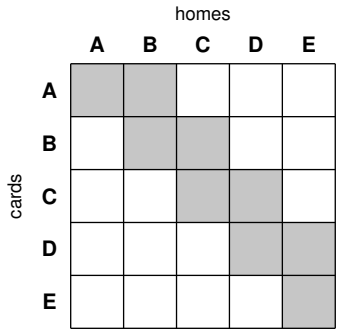


Problem 11, four cards

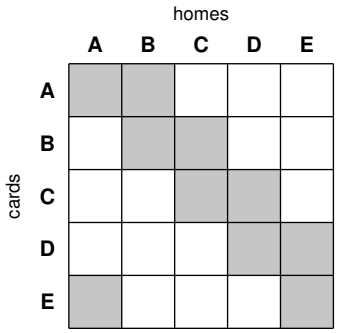
extra grids



Problem 13: Count the rows that obey each of these five-card grids without listing them.



Problem 10, five cards



Problem 11, five cards

extra grids

